Note 6 White Noise Tests

In time series analysis, it is important to know if a series is stationary or not, which can be determined by the series plot or some statistical tests.
Once we have decided a series is stationary, we also would like to know if this stationary series is white noise or not. Because for a white noise series, no other modeling is necessary except estimating the variance of the series. But for series that are not white noise, we need to model the dependence structure of the series. Moreover, white noise tests can also be used in residual diagnostics.
In this note, we briefly introduce several statical tests for white noise.

1 Sample Autocorrelation Functions

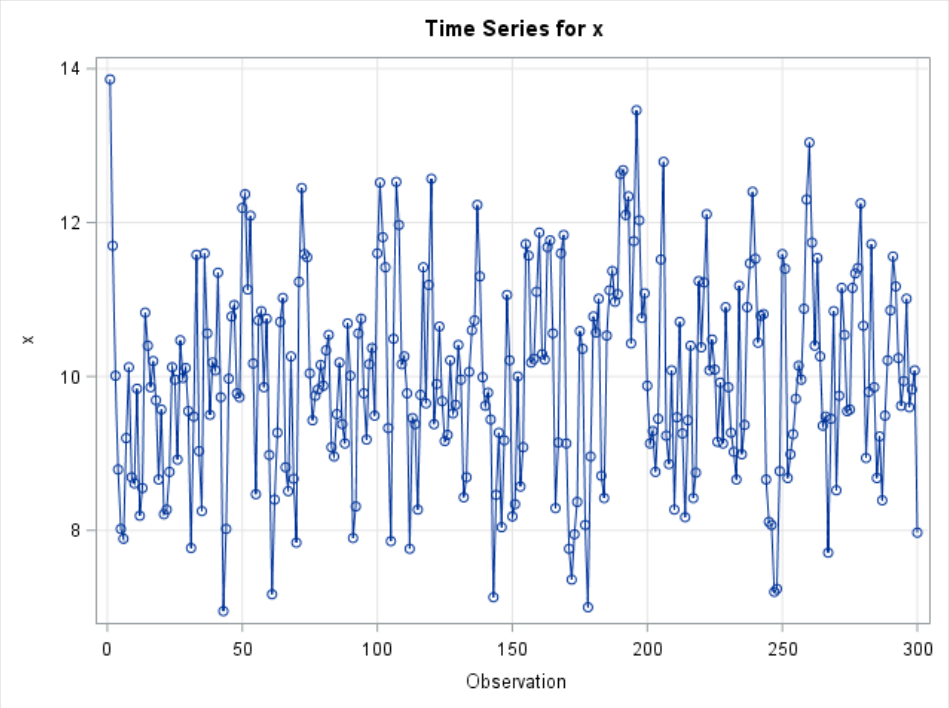
Let $\{X_t\}$ be a stationary time series of length n, recall the sample autocorrelation function is

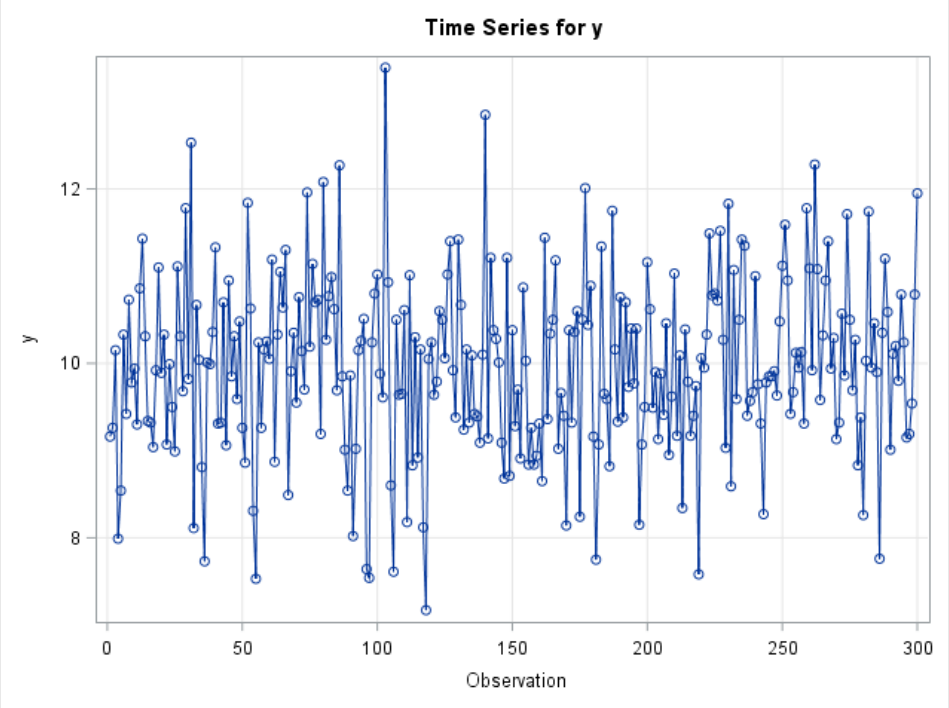
$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)} = \frac{\sum_{t=1}^{n-h}(X_t - \bar{X})(X_{t+h} - \bar{X})}{\sum_{t=1}^{n-h}(X_t - \bar{X})^2}$$

If $\{X_t\}$ is white noise of length n, for n large enough,

$$\hat{\rho}(h) \sim N\left(0, \frac{1}{n}\right) \text{ approximately.}$$

Example 1.1. Below are the plots of the two series, one is white noise and the other one is not. Identify the white noise.





2 Ljung-Box Test

For a stationary time series $\{X_t\}$, the Ljung-Box test is an overall test on the autocorrelation functions up to a specific lag h. The hypotheses are usually stated as follow:

$$H_0 : \rho(1) = \rho(2) = \cdots = \rho(h) = 0 \text{ (The data are independently distributed)}$$

$$H_1 : \text{at least one of } \rho(i) \neq 0 \text{ (The data are not independently distributed)}$$

The lag h is usually a relatively large number. SAS show the result for $h = 6, 12, 18, 24$. The test statistic is

$$Q = n(n+2) \sum_{j=1}^{h} \frac{\hat{\rho}(j)^2}{n-j}$$

Under H_0, the statistic Q asymptotically follows a $\chi^2(h)$.